Assignment - 01

Tutorial Sheet - 01

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DBMS

- Sec-A (2×5 = 10)

1. What do you understand by DBMS ?

Ans DBMS stands for database Management system. It is a software system that enables users to define, create, maintain, and manage database. DBMS provides a systematic and organized way to store, retrive and manipulate data. Example of Popular database management system include MySQL; oracle database, MS SQL server.

2. What is E-R model ?

Ans The E-R (Entity Relationship) model is a conceptual data model used to describe the relationships between different entities in a database. It represent's real-world entities as "entities" and the relationships between then as "relationships". Entities are represented as rectangles and relationships are represented as diamonds in an E-R diagram.

3. Explain the importance of transaction management.

Ans Transaction management is crucial in ensuring the integrity, consistency, and reliability of data within a database system. The importance of transaction management can be summarized as follows:

- Atomicity :- Transaction should be atomic, meaning they should either be complete in full or not at all.
- Consistency :- Transaction should maintain the consistency of the database, adhering to predefined constraints and integrity rules.
- Durability :- once a transaction is commited, its changes should be permanent and survive system failures.

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4. Describe the ~~key~~ concept of keys in DBMS.

Ans Key in DBMS are attributes or combinations of attributes that uniquely identify rows (records) within a database table. They ensure data integrity and provide a means to establish relationships betn. tables. There are several types of keys.

- Primary key - A P.K is a unique identifier for each record in a table.
- Foreign key - A F.K is a set of column in one table that refers to the primary key in another table.
- Candidate key - A candidate key is a set of attributes that can uniquely identify each record in a table.
- Composite key - A composite key is a key that consists of multiple columns to uniquely identify a record.

5. Explain the diffn. betn. data and information.

Ans

- Data: refers to rows facts, figures, or symbols that have little or no meaning on their own. it is unprocessed and lacks context; for eg:- a list of no., characters, or words.

- information, on other hand, is processed and organized data that has been interpreted within a context to make it meaningful and useful for decision-making or understanding. information provides insights, knowledge or understanding derived from analyzing and interpreting data. for ex: a summary report, chart, or analysis based on the list of no. would be considered information.

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• Sec-B (3×7=21)

Q 6. What are the different DBMS language ? Also explain their commands.

Ans Database Management system (DBMS) utilize various specialized languages to execute different tasks. Here's a breakdown of the Primary categories of DBMS lang. along with their typical commands:

(i) Data Definition language (DDL):- it is used to Define the structures and schema of the database and its Objects (table, columns, etc).

DDL Commands :-

- Create : Create new database objects.
- Alter : Modifies the structure of existing database object.
- Drop : Delete database table.
- Truncate : Deletes all data from a table while retaining the structure.

(ii) Data manipulation language (DML):-

- Function - Enables manipulation of data within existing database objects.

DML Commands :-

- SELECT : Retrieves data from one or more tables.
- INSERT : Adds new rows of data into a table.
- UPDATE : Modifies existing data within a table.
- Delete : Removes rows from a table.

(iii) Data Control language(DCL):-

- Function - manages access rights and Permissions for users and roles.

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DCL Commands :-

- GRANT : Assigns privileges to users
- Revoke : Takes away previously granted Privileges.

(iv) Transaction Control Language (TCL):-

- Function - Controls groups of DML statements to ensure that transactions are atomic and Consistent.

TCL Commands:-

- COMMIT : Permanently saves changes made within a transaction.
- Rollback: undoes changes since the beginning of a transaction.

Q7. Explain the architecture of DBMS.

Ans. The term "database architecture" refers to the structural design and methodology of a database system, which forms the core of a database Management system (DBMS). This architecture dictates how data is store, organized, and retrieved, playing a crucial role in the efficiency and effectiveness of data management.

- The three -Tier (or Three - Schema) Arhitecture :-

The most common type of DBMS architecture is the three - tier architecture. it's designed to separate the presentation of data, the application logic, and the physical data storage, giving you a modular and flexible system. Here's the breakdown :-

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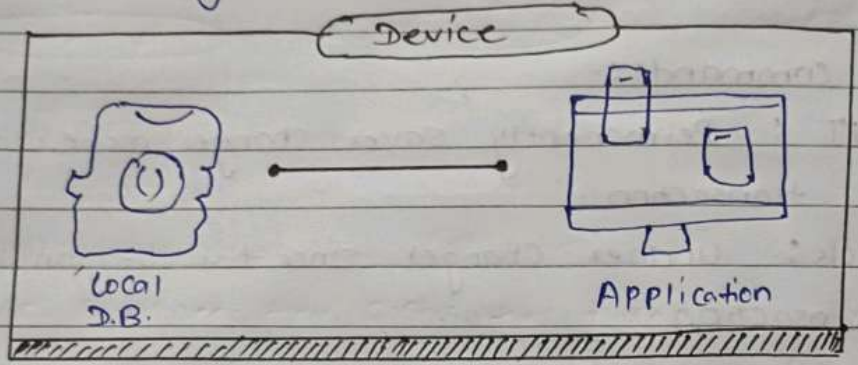
(i) One-tier architecture :- in one-tier architecture. the database, user interface, and application logic all reside on the same machine or server. it's typically used for small-scale applications where simplicity and cost effectiveness are priorities. Because there are no network delays involved, this type of tier architecture is generally a fast way to access data.



(on a single-tier application, the application and database reside on the same device.)

(ii) Two-tier architecture :- Two tier architecture consists of multiple clients connecting directly to the database. This tier architecture is also known as client-server architecture.

This tier architecture used to be more common when a desktop application would connect to a single database hosted on an on-premise database server - for ex:- an in-house customer relationship management (CRM) that connects to an Access database.

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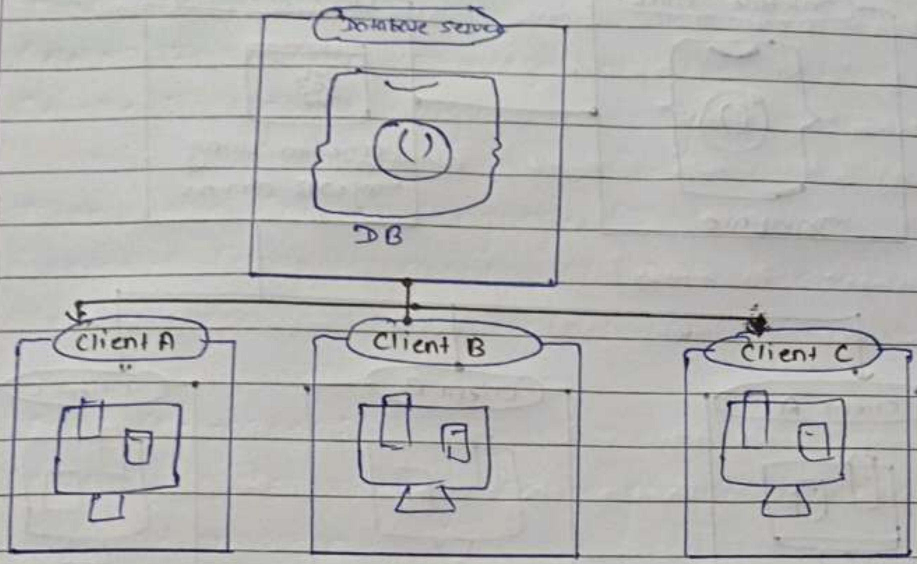
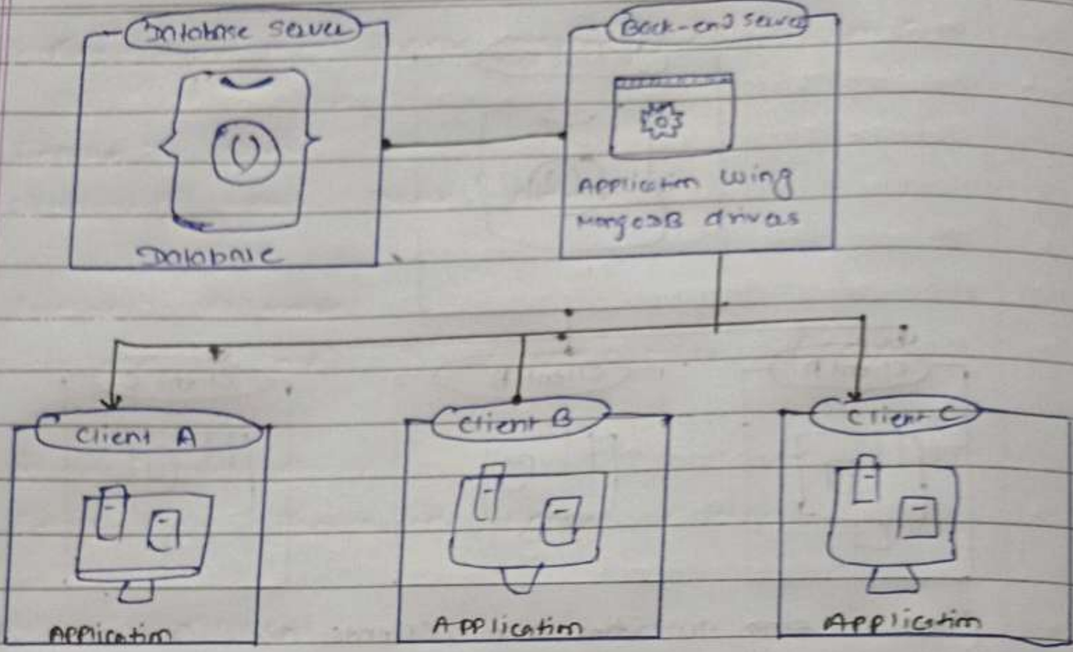


(on a ~~sing~~ two-tier arch., Clients are connecting directly to a database.)

(iii) Three-tier architecture :-

Most Modern Web applications use a three-tier architecture. in this architecture, the clients connects to a back end, which in turn connects to the database. using this approach has many benefits:

- Security: keeping the dB connection open to a single back end reduces the risks of being haked.
- Scalability :- because each layer operates independently, it is easier to scale parts of the application.
- faster deployment : Having multiple tiers makes it easier to have a separation of concerns and to follow cloud-native best practices, incluwing better continuous delivery processes.



(in a three-tier arch., the information betn. the database and the clients is relayed by a back-end server).

- An example of this type of arch. would be a React app that connects to a Node.js back end. The Node.js back end processes the requests and fetches the necessary information from a database such as MongoDB Atlas, using the native driver.

Q8. What are the key difference betn. file system and DBMS?

Ans

File system	DBMS
• It is used to manage and organise the files stored in the hard disk of the computer system.	• A software to store and retrieve the user's data.
• Query processing is not so efficient.	• Query processing is efficient.
• Data consistency is low.	• Due to the process of normalisation the data consistency is high.
• Less security.	• Supports more consistency mechanisms.
• Does not support crash recovery.	Crash recovery mechanism is highly supported.

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Sec-C (3x11 = 33)

Q9. How do we represent E-R diagrams in DBMS? Explain the E-R diagram of a bank management system?

Ans in dbms, an Entity-Relationship (E-R) diagram is a visual representation of the data model that describes how data is related to each other within a system. it consists of entities (which represent real-world objects or concepts), attributes (properties of entities), and relationships (connections between entities).

Here's how we represent an E-R diagram in DBMS.

(i) Entities :- These are represented by rectangles in the diagram. Each entity represents a distinct object or concept in the system being modeled. for example, in a bank management system, entities might include "customer", "Account", and "Transaction", etc.

(ii) Attributes: These are characteristics or properties of entities and are depicted as ovals connected to the corresponding entity rectangle. for instance, attributes of a "customer" entity might include "Customer ID", "Name", "Address", etc.

(iii) Relationships: These illustrate how entities are related to each other. Relationships are depicted by diamond shapes connecting the related entities. They show how two entities share information with each other. for instance, in a bank management system, a "customer" entity may have a relation with an "Account" entity.

(iv) Primary keys: These are unique identifiers for each entity and are often underlined in the diagram. They help uniquely identify each record in the database.

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- E-R diagrams of Bank Management system :-

- Entities :- Bank
  Attributes : Name, code, Address,

- Entities : Customer
  Attributes : Name, ID, Phone number, Address

- Entity : Branch
  Attribute : Branch-id, name, Address

- Entity : Account
  Attribute : Account-number, type, balance

- Entity : Loan
  Attribute : Loan-id, Loan-type, Amount.

- Relationship are :

  Bank has Branches => 1 : N

  Branch Maintain Accounts => 1 : N

  Branch Offers Loans => 1 : N

  Accounts held by coustomers => M : N

  Loan availd by Customer => M : N

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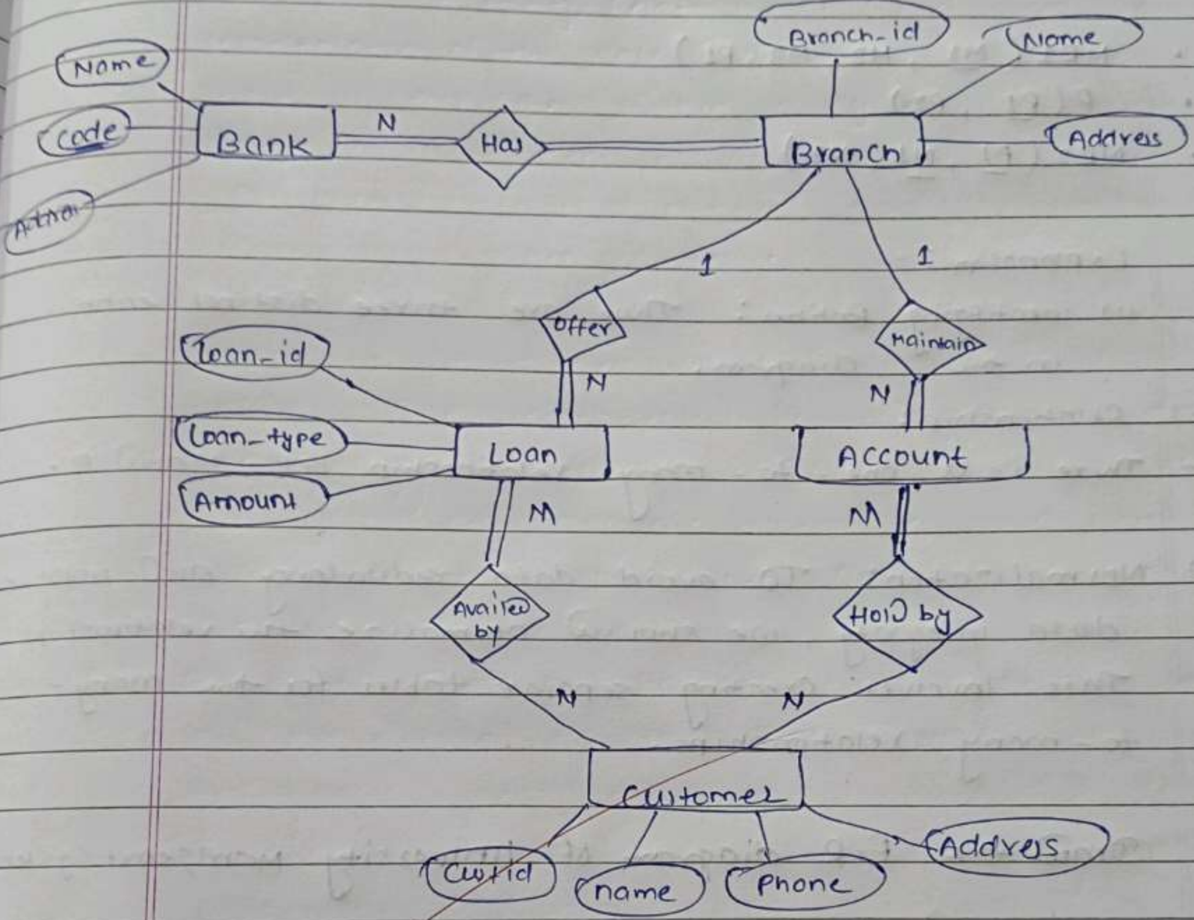
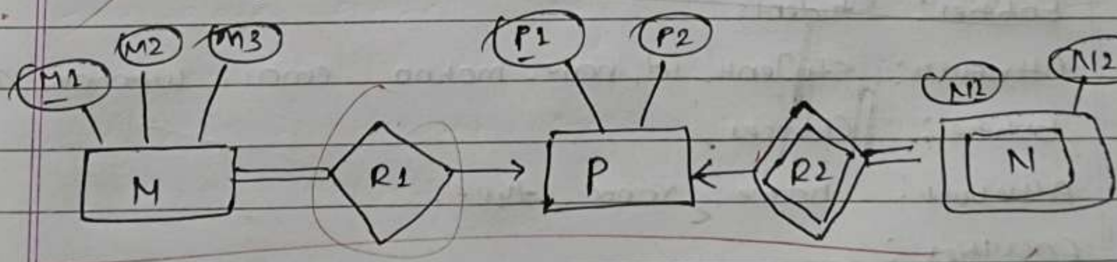
• Diagram :-



Q10. Find the minimum no. of tables required for the following ER diagram in relational model -

3.



Ans In the above E-R diagram there are total 3 tables will be required.

let's explain

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Applying the rules, minimum 3 tables will be required -

- MR1 (M1, M2, M3, P1)
- P (P1, P2)
- NR2 (P1, N1, N2)

- Explanation :-
(i) identifying Entities: There are three distinct entities in the diagram.
(ii) Relationship:
- There is a one-to-many relationship betn. M and P.
-
(iii) Normalization: To avoid data redundancy and improve data integrity, we should normalize the relationships. This involves creating separate tables for the many-to-many relationships.

Q11. Draw the E-R diagram of University management system

Ans. • Entities: College
Attributes: College-id, name, address
• Entities: Students
Attribute: student_id, name, mobno, email, username, address
• Entities: Classes
Attribute: name, room, type,
• Faculties:
Attribute: id, name, room, type, mob-no.
• Entity: Course
Attribute: id, name, type, year

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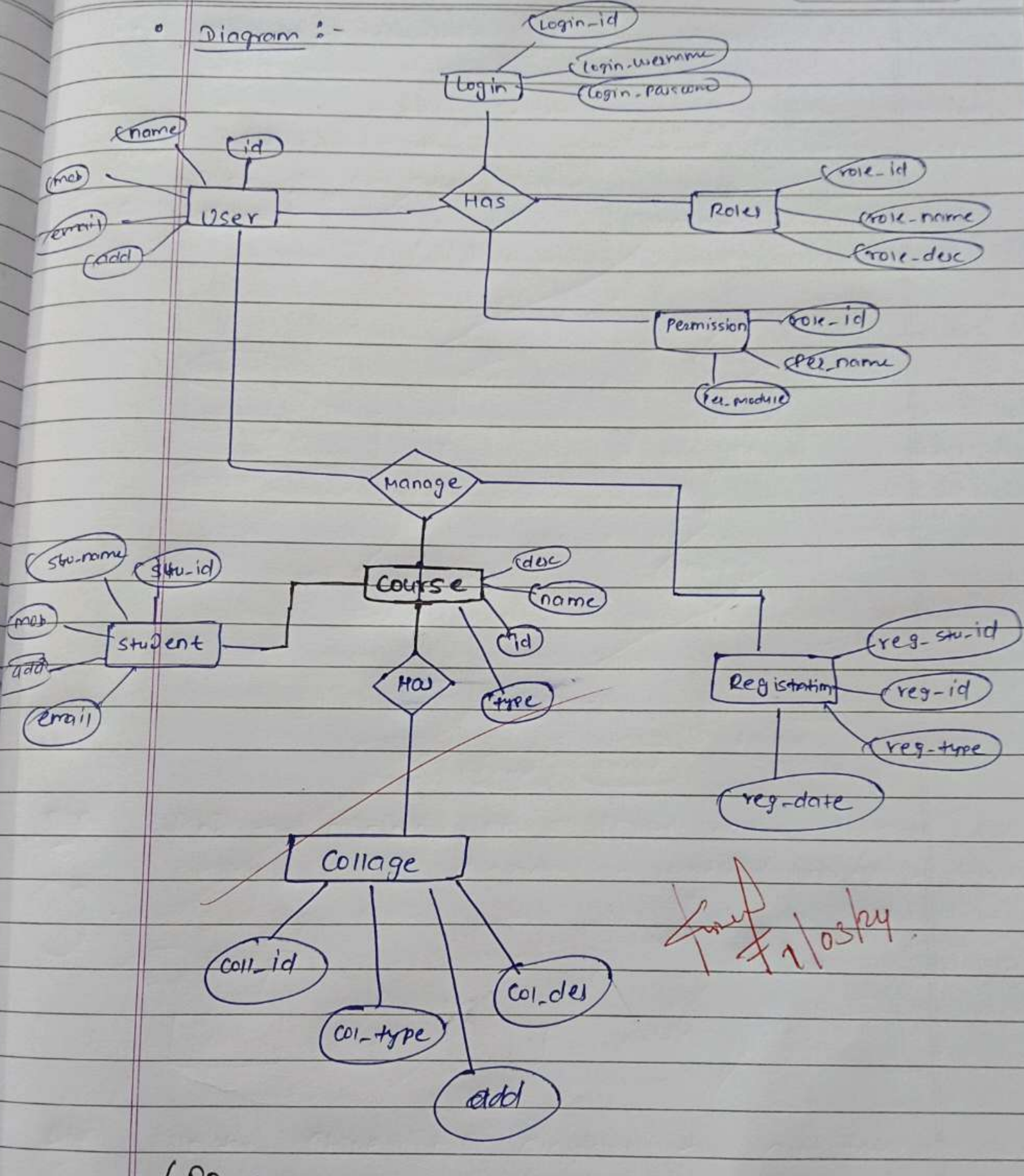
• Diagram :-



(ER diagram for University Management system)

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